

OPARINO, Russian Federation

WMO: 270830

Lat: 59.850N

Lon: 48.283E

Elev: 180

StdP: 99.19

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.5	-28.1	-34.0	0.2	-31.5	-30.7	0.2	-28.1	6.3	-5.8	5.7	-6.4	0.8	10	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	28.1	19.7	26.2	18.6	24.4	17.6	20.8	26.3	19.8	24.6	18.8	23.0	2.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.0	23.2	18.0	13.2	22.1	17.0	12.4	21.2	60.8	26.6	57.3	24.7	54.0	22.9	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.3	4.6	-35.3	30.7	4.4	1.7	-38.5	31.9	-41.1	32.9	-43.5	33.9	-46.8	35.1		
(5)			-35.4	22.6	4.4	1.7	-38.5	23.9	-41.1	24.9	-43.6	25.8	-46.8	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-12.7	-11.3	-4.7	3.0	9.9	14.6	17.3	14.5	9.1	2.5	-4.8	-10.1
	DBStd	11.79	7.90	7.53	5.05	4.88	5.28	4.40	3.80	4.03	4.03	4.35	6.15	7.97
	HDD10.0	3425	705	595	455	215	71	9	1	7	63	236	445	623
	HDD18.3	5903	963	828	713	460	266	128	66	133	277	492	695	882
	CDD10.0	628	0	0	0	5	66	147	226	145	37	3	0	0
	CDD18.3	64	0	0	0	0	3	15	33	13	1	0	0	0
	CDH23.3	619	0	0	0	2	54	151	301	108	4	0	0	0
	CDH26.7	130	0	0	0	0	8	31	64	27	0	0	0	0
Wind	WSAvg	2.0	2.0	2.1	2.3	2.0	2.1	1.9	1.6	1.7	1.9	2.3	2.2	2.2
Precipitation	PrecAvg	669	41	32	38	34	57	85	77	84	59	61	53	48
	PrecMax	863	65	78	78	64	138	154	175	182	122	109	106	84
	PrecMin	502	18	9	6	3	15	17	11	26	16	24	15	30
	PrecStd	93	11	17	17	15	26	37	43	31	27	26	21	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.9	8.9	21.0	27.8	30.2	30.8	30.0	23.2	16.8	8.0	3.5
		MCWB	0.1	0.7	3.9	12.4	17.4	20.8	21.0	21.0	16.0	10.0	7.1	3.2
	2%	DB	0.2	0.7	5.6	16.5	24.6	27.0	28.5	26.2	20.0	12.7	4.8	1.5
		MCWB	-0.3	0.1	2.0	9.3	15.1	18.8	20.2	19.6	14.9	9.5	4.0	1.2
	5%	DB	-0.9	-0.1	3.4	13.4	21.9	24.9	26.6	23.6	17.7	10.4	3.0	0.5
		MCWB	-1.4	-0.6	0.7	7.2	13.4	17.5	19.3	17.8	13.6	8.4	2.4	0.1
	10%	DB	-2.6	-1.4	1.8	10.9	19.4	22.8	24.8	21.1	15.5	8.6	1.6	-0.5
		MCWB	-3.0	-2.1	0.0	5.9	12.5	16.6	18.4	16.6	12.6	7.2	1.1	-0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	1.1	4.5	12.7	18.7	21.9	22.1	21.6	17.0	12.1	7.1	3.2
		MCDB	0.7	1.4	8.0	20.6	25.6	28.5	28.1	28.1	20.8	15.4	7.9	3.4
	2%	WB	-0.2	0.3	2.6	9.8	16.4	20.2	21.0	20.1	15.5	10.0	4.1	1.2
		MCDB	0.2	0.7	4.7	15.5	22.3	25.2	26.5	24.9	18.9	12.0	4.6	1.4
	5%	WB	-1.4	-0.6	1.2	8.0	14.7	18.8	20.1	18.8	14.2	8.5	2.5	0.1
		MCDB	-0.9	-0.1	2.8	12.4	20.0	22.8	25.1	22.4	17.1	10.1	2.9	0.3
	10%	WB	-3.0	-2.0	0.2	6.3	13.2	17.4	19.2	17.4	12.9	7.2	1.2	-0.9
		MCDB	-2.6	-1.5	1.6	10.3	18.4	21.5	23.8	20.3	15.2	8.5	1.6	-0.5
Mean Daily Temperature Range	5% DB	MDBR	6.0	7.2	8.0	9.3	10.7	9.9	9.9	8.7	7.5	4.6	4.0	5.1
		MCDBR	5.1	5.5	9.9	14.2	14.9	13.2	13.1	12.1	11.5	8.1	4.5	3.4
		MCWBR	5.2	5.3	6.9	8.1	7.1	6.3	5.8	5.7	6.6	5.4	4.3	3.4
	5% WB	MCDBR	5.0	5.3	7.5	12.6	12.9	11.4	11.6	11.1	9.8	7.3	4.2	3.5
		MCWBR	5.2	5.3	5.6	7.8	6.9	6.0	5.7	5.8	6.1	5.3	4.1	3.6
Clear-Sky Solar Irradiance	taub	0.265	0.303	0.340	0.376	0.361	0.359	0.391	0.384	0.348	0.325	0.286	0.246	
	taud	2.034	2.065	2.046	2.158	2.418	2.449	2.363	2.364	2.446	2.372	2.148	1.987	
	Ebn at Noon	483	671	766	815	867	874	831	806	779	668	462	351	
	Edh at Noon	48	82	114	123	103	102	108	101	79	61	43	33	
All-Sky Solar Radiation	RadAvg	0.35	1.05	2.26	3.73	4.97	5.27	5.33	3.83	2.19	0.93	0.32	0.18	
	RadStd	0.05	0.26	0.35	0.47	0.49	0.45	0.52	0.40	0.29	0.12	0.06	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.59	N/A	N/A	N/A	N/A	+0.63	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page